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Assignment-2 (Graph theory & applications)

Q2:- let H be a new graph formed by combining edges of M & N

Because M & N are Matchings, no vertex in H connects to more than 2 edges. This means H only consists of separate paths & cycles.

∀ cycle must alternate b/w M & N edges so it have same no. of edges from both.

∵ $|M| > |N|$ this extra edge of $|M|$ must be in a path so there must be atleast one alternating path (call it P) that starts & ends with an M edge.

Now create M' & N' by swapping the edges only inside path P :- give its M edges to N' & its N edges to M'

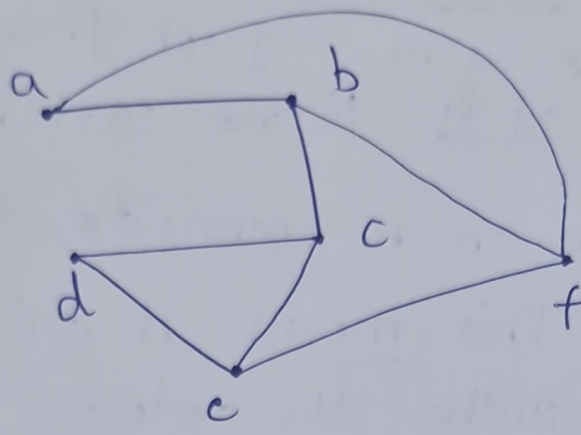
so:- $|N'| = |N| + 1$ (as N' gains that extra M edge)

$|M'| = |M| - 1$ (as M loses exactly 1 edge)

& and total edges stays same

$(M' \cup N' = M \cup N)$

Ques 3:-



Start with empty independent set $I = \phi$

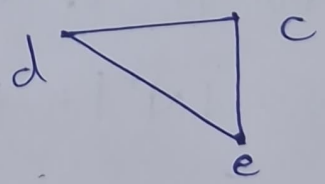
& $V = \{a, b, c, d, e, f\}$

step 1:- pick 'a' & add to I

so $I = \{a\}$

remove a & all its neighbours from V

$V = \{c, d, e\}$



step 2:- pick 'c' add to I

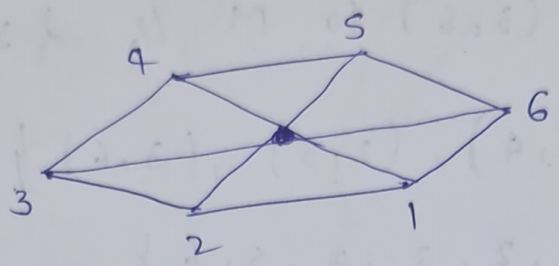
$I = \{a, c\}$

remove c & all its neighbors ~~from~~ from V

$V = \phi$

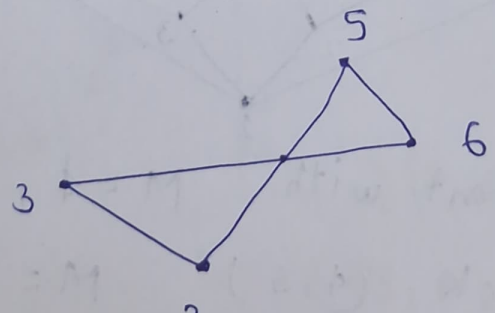
as $V = \phi$ algo terminates final maximal $I = \{a, c\}$

Q 4 :-



step 1 :- $M = \phi$ (starting with empty matching & vertex cover)
$K = \phi$

step 2 :- choose edge (1, 4) add to
$M \Rightarrow M = \{(1,4)\}$ & add
these vertices to K
$K = \{1, 4\}$ & delete $\{1, 4\}$
from G



step 3 :- choose (2, 5) ~~M =~~ · add to M
$M = \{(1,4), (2,5)\}$ & add vertices
to K $\Rightarrow$ $K = \{1, 2, 4, 5\}$ ~~G~~ = delete 2, 5
from G



Step 4:- choose (3,6) delete it from G & add (3,6) to M & {3,6} to K

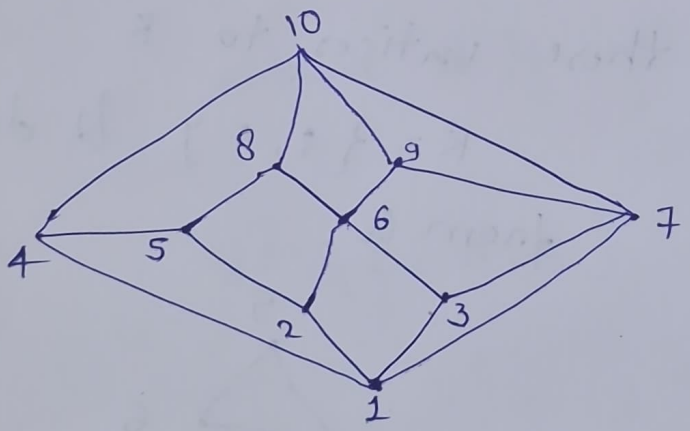
$M = \{(1,4), (2,5), (3,6)\}$

$K = \{1, 2, 3, 4, 5, 6\}$

no valid edge in G so algo terminates.

Maximal Matching $M = \{(1,2), (4,5), (3,6)\}$

Vertex cover $C = \{1, 2, 3, 4, 5, 6\}$

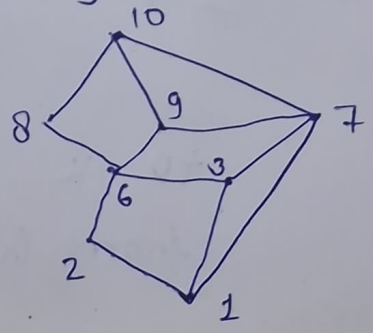


step 1:- start with $M = \phi$, $K = \phi$

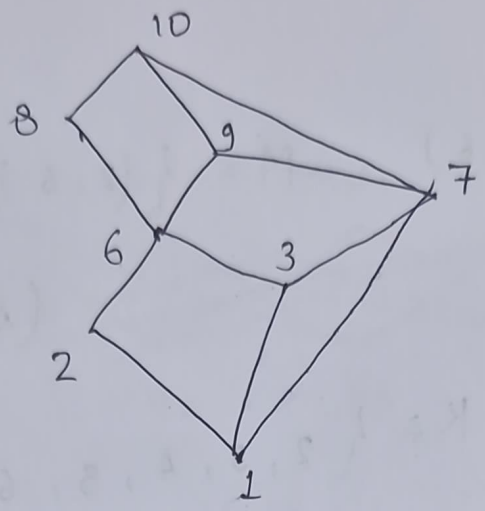
step 2: pick (4,5) $M = \{(4,5)\}$

$\{4,5\}$ $K = \{4,5\}$

delete from G



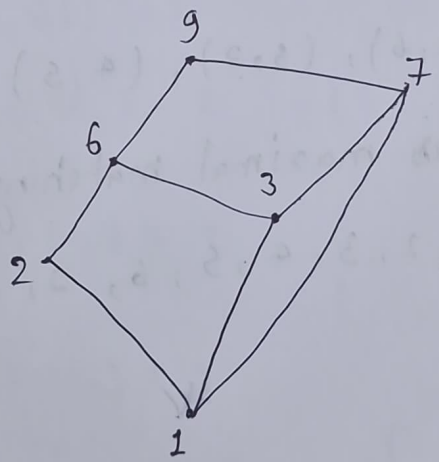
Q4



Step 3:- pick (8,10) $M = \{(4,5), (8,10)\}$

$K = \{4, 5, 8, 10\}$

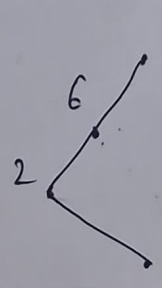
delete {8,10} from G



step 4:- pick → (3,7) $M = \{(4,5), (8,10), (3,7)\}$

$K = \{3, 4, 5, 7, 8, 10\}$

delete {3,7} from G



Steps:

pick (2,6) $M = \{(2,6), (3,7), (4,5), (8,10)\}$

$K = \{2, 3, 4, 5, 6, 7, 8, 10\}$

delete $\{2, 6\}$ from G

i g

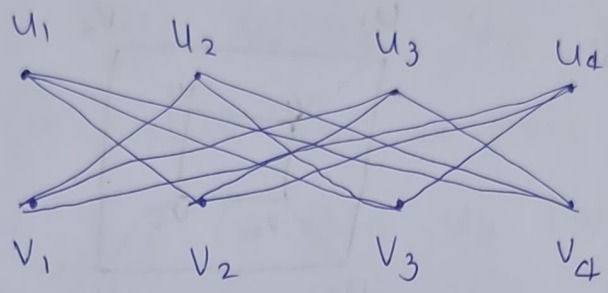
as no valid edges available algo. terminates

$M = \{(2,6), (3,7), (4,5), (8,10)\}$

↳ is maximal matching

& $K = \{2, 3, 4, 5, 6, 7, 8, 10\}$ is vertex cover

Qs:-



valid order:- $S = \{u_1, v_1, u_2, v_2, u_3, v_3, u_4, v_4\}$

testing greedy algo on S

$c(u_1) = 1$

$c(v_1) = 1$

$c(u_2) = 2$ ($\because u_2$ is connected to v_1 & v_1 has colon 1)

$c(v_2) = 2$ ($\because v_2$ is connected to u_1 & $c(u_1) = 1$)

$c(u_3) = 3$ ($\because N(u_3) = \{v_1, v_2, v_4\}$ & $c(v_1) = 1$ & $c(v_2) = 2$)

$c(v_3) = 3$ ($\because N(v_3) = \{u_1, u_2, u_4\}$ $c(u_1) = 1$ & $c(u_2) = 2$)

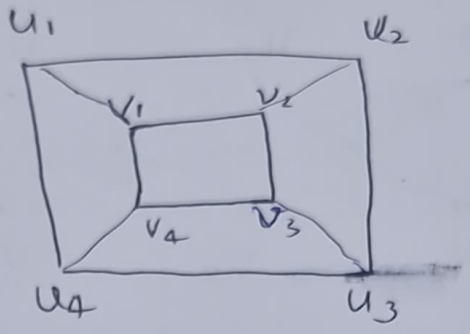
$c(u_4) = 4$ $\because$ all neighbours has 1, 2, 3

$c(v_4) = 4$ $\because$ all neighbours has 1, 2, 3

So we got 4 colon by greedy algo.

Q6:-

Q5:-



6(a):- $S_1 = \{ u_1, v_2, v_4, u_3, u_2, u_4, v_1, v_3 \}$

as A = $\{u_1, v_2, u_3, v_4\}$

B = $\{u_2, v_4, v_1, v_3\}$

AS A ; B ~~is~~ pantition is bipantite. no edge b/w group A vertices & no edge b/w ~~p~~ group B vertices so needed colon

1 + 1 = 2

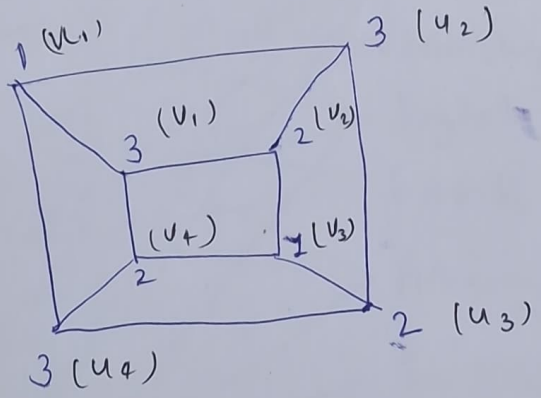
6(b):- suppose we want to give vertex v_1 the 3rd colon so we keep v_3 first in seqn S_2 so v_3 get colon '1' now keep v_2 & v_4 they get colon 2 now only remaining uncolored neighbors of v_1 is u_1 give it colon 1 by placing it at th v_4 now if we put v_1 in seqn it will get

colon 3.

so till $\{V_3, V_2, V_4, U_1, V_1\}$

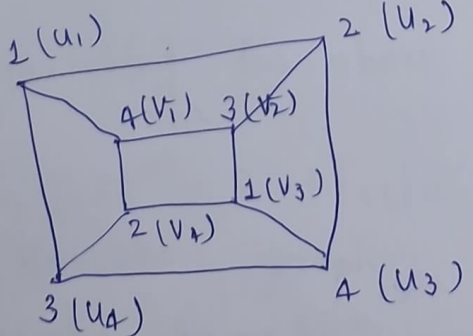
we got ~~col~~ 3 colons now we place V_2 (it will get colon 3) & then U_3 (will get colon 2) & U_4 (get colon 3)

$$S_2 = \{V_3, V_2, V_4, U_1, V_1, U_2, U_3, U_4\}$$

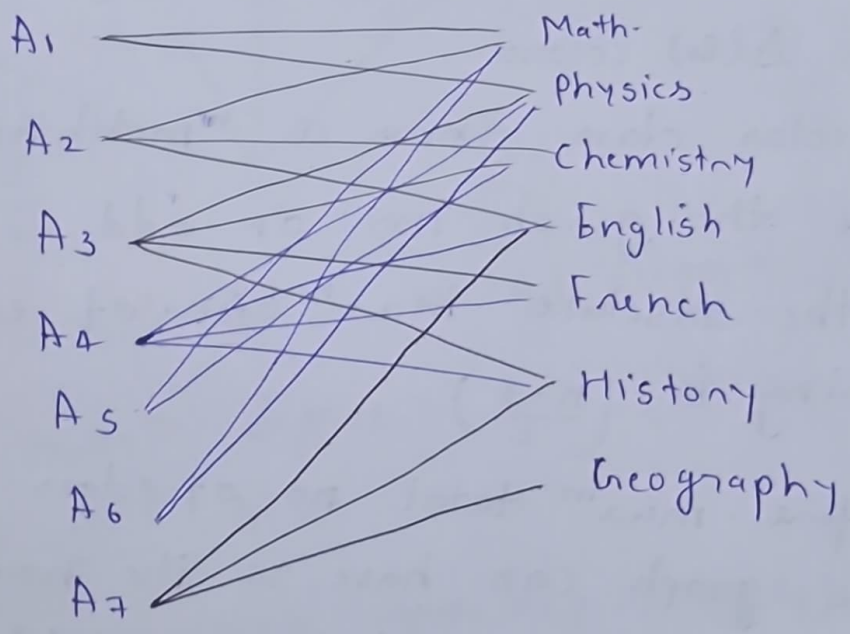


6 (c) :- for 4 colon :-

$$S_3 = \{U_1, V_3, U_2, V_4, U_4, V_2, U_3, V_1\}$$



Ans 8:- modeling it as graph problem making teachers & subjects as vertices & edges if teacher ~~teach~~ knows that subject bipartite graph as no edges b/w teacher themselves & no edges b/w subjects themselves



now to find max^m no. of suitably qualified teachers we have to find the maximal perfect matching for above graph.

we get max^m 7 teaches a valid perfect matching for above graph.-

$A_7 \rightarrow$ Geography, $A_1 \rightarrow$ Math, $A_6 \rightarrow$ Physics, $A_5 \rightarrow$ Chemistry, $A_2 \rightarrow$ English, $A_3 \rightarrow$ French, $A_4 \rightarrow$ History

colon 6:- $\{(v_4, v_5), (v_3, v_7), \ldots$

Ques 9 :- given $2m > (n-1)\,\Delta(G)$

According to vizing's theorem, the edge chromatic number of any graph $\chi_1(G)$ can only be exactly $\Delta(G)$ on $\Delta(G)+1$

Let's assume for contradiction $\chi_1(G) = \Delta(G)$
this means we can colon all edges using exactly $\Delta(G)$ colons.
each colon class forms a "matching"
because the graph has an odd no. of vertices (n) the absolute maxm no. of edges in any matching is $\left(\frac{n-1}{2}\right)$
therefore, maxm total no. of edges m the entire graph can have is the no. of colons multiplied by the maxm matching size

$$m \leq \Delta G \times \frac{(n-1)}{2}$$

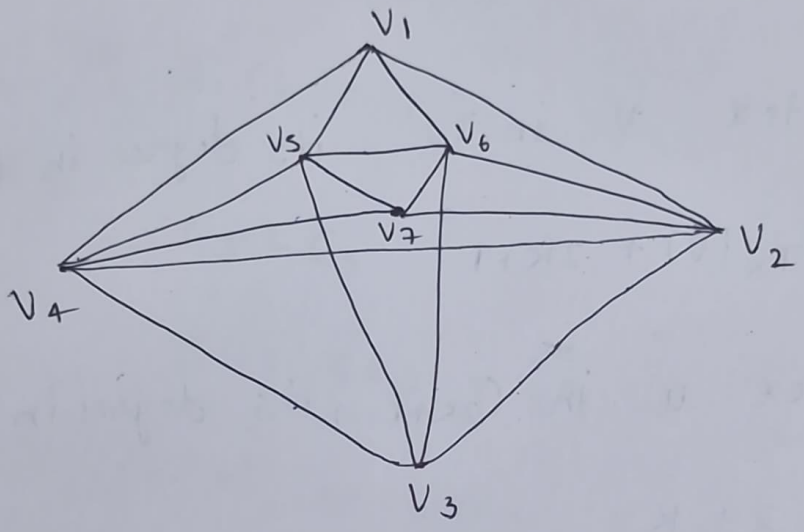
$$2m \leq \Delta G\,(n-1)$$

but it was given $2m > \Delta G\,(n-1)$ hence contradiction.

$\therefore$ $\chi_1(G)$ can't be $\Delta(G)$ it must be only other option allowed by vizing's theorem $\chi_1(G) = \Delta(G)+1$

$\chi_1(G) - 1 = \Delta(G)$ Proved

Soln 9 (part-2) :-



n = 7 odd no. of edges = 17

max^m degree = 5 (of V_5, V_6, V_4, V_2)

$\Delta(G) = 5$

checking $2m \geq (n-1)(\Delta G)$

$2 \times 17 > 6 \times 5$

$34 > 30$

so as it satisfies so $\chi_1(G) = 5+1 = 6$

colon 1:- $\{(V_1, V_2), (V_3, V_5), (V_6, V_7)\}$

colon 2:- $\{(V_1, V_4), (V_2, V_7), (V_5, V_6)\}$

colon 3:- $\{(V_1, V_5), (V_2, V_6), (V_3, V_4)\}$

colon 4:- $\{(V_1, V_6), (V_2, V_3), (V_4, V_7)\}$

colon 5:- $\{(V_2, V_4), (V_3, V_6), (V_5, V_7)\}$

colon 6:- $\{(V_4, V_5), (V_3, V_7)\}$

soln 7 :-

(a)

$G = T_K + C_{2K+1}$

for any vetex V in T_K its degree in G

$= d_{T_K}(V) + 2K+1$

for any vertex u in C_{2K+1}, its degree in G

$= 2 + K$

for $K \geq 2$ $(2K+1) > (K+2)$ this means the vertices with the absolute maxm degree in G are completely contained within T_K.

$\because \quad \Delta(G) = \Delta(T_K) + 2K+1$

By vizing's theorem the $\chi_1(G)$ is ΔG or $\Delta G + 1$

we look at the subgraph ~~$\Delta G + 1$~~ formed only by the vertices of maxm degree. Since all these maxm degree vertices live exclusively inside T_K, the subgraph they induce is a forrest.

By Fournin's theorem, if the subgraph induced

Soln 7:- (a)

By all max^m degree vertices is a forest, the graph is "class 1".

so $\chi_1(G) = \Delta(T_k) + 2k + 1$

7 (b):- $G = C_5 + C_5$

$\Delta G = 5 + 2 = 7$

so $\chi_1(G)$ be either 7 or 8

let's assume $\chi_1(G) = 7$

total no. of edges in G is $\frac{(10 \times 7)}{2} = 35$

if we colon it using 7 colon, each colon should appear exactly $\frac{35}{7} = 5$, 5 times

this means 1 colon class forms a perfect matching across 10 vertices.

K_{10} can be split into 9 perfect matching

if G uses 7 of these perfect matchings

$\bar{G}$ should be formed by $9 - 7 = 2$ perfect matchings.

$\bar{G} = \bar{C_5} \cup \bar{C_5}$ as $\bar{C_5}$ is C_5

$\bar{G} = C_5 \cup C_5$

but an odd cycle like C_5 can't be mathematically contain a perfect matching, thus $\bar{G}$ can't be made of 2 perfect matching. Hence contradiction

$\therefore \chi_1(G)$ can't be 7 so it must be 8.

$\chi_1(C_5 + C_5) = 8$

Ans 7 (c) :-

$C_{2s} + C_{2t}$

$\Delta G = \max(2s+2, 2t+2)$

now edges ~~can~~ are here 3 types

↳ ① b/w C_{2s}

② b/w C_{2t}

③ ~~on~~ b/w C_{2s} to C_{2t}

now C_{2s} is even cycle so it need 2 colon to colon

edges b/w C_{2t} also need 2 colons as even cycle we can use colon given to edges b/w C_{2s} so total colon used = 2

now for edges b/w C_{2s} to C_{2t} it is like $K_{2s,2t}$ these edges can be represented by this

Ans 7(c):-

now as edge chromatic no. of complete bipartite graph is

$$\max(2s, 2t)$$

so total colors $2 + \max(2s, 2t)$

$$\chi_1(C_{2s} + C_{2t}) = \max(2s+2, 2t+2)$$

$$= 6$$

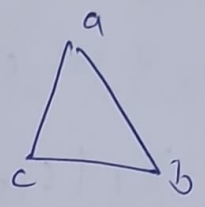
Ans

Ans 1 (a) :-

$$\beta_0(G) + \beta_0(G^c) \geq \alpha_0(G) + \alpha_0(G^c)$$

this inequality is wrong

counter example K_3



$$\alpha_0(K_3) = 1$$

$$\beta_0(K_3) = n - 1 = 3 - 1 = 2$$

K_3^c =



$$\alpha_0(K_3^c) = 3$$

$$\beta_0(K_3^c) = 3 - 3 = 0$$

now $\Rightarrow 2 + 0 \geq 3 + 1$

$2 \not\geq 4$

so inequality not holds so it is wrong

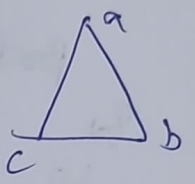
this inequality is not true for all graphs

Ans 1(b):-

$$\beta_0(G) \geq \omega(G)$$

it is also false

counter example = K_3



Here max^m clique size = 3

$\beta_0(G)$ = 2 (ab & bc covers all vertices)

$2 \geq 3$ not holds so

$\beta_0(G) \geq \omega(G)$ is not true for all graphs.

Ans 1(c):- $\beta_0(G) \geq \delta(G)$

it is true

proof:- let K be min^m vertex cover of graph G, meaning $|K| = \beta_0(G)$

if all vertices of G is in (K) then

$$|K| = n \geq \delta(G)$$

so suppose a vertex v not in K so as K is valid vertex cover so to cover edges incident on v must be covered by its

A

neighbon and they must be in k

as neighbon of $v = d(v)$

& $d(v) \geq \delta(G)$

so

$\beta_0(G)$ $|k| \geq d(v) \geq \delta(G)$

& $\beta_0(G) = |k|$

so $\beta_0(G) \geq \delta(G)$

Proved

Ques 10(a) :-

$n = 10$, 3-regular so $d(v_i) = 3$ for $i = 1, \ldots n$

m = no. of edges f = no. of faces

$$2m = \sum_{i=1}^{n} \deg(v_i)$$

$$2m = 3 \times n$$

$$\boxed{m = 15}$$

Proving graph is connected:-

pick a vertex say (u) as $d(u) = 3$

so u is connected to 3 neighbors

∵ girth is 5 graph can't have 3-or 4 cycles

so these 3 neighbors can't share edges & also can't share neighbors

so total vertices = $1 + 3 + 3 \times 2 = 10$

so without breaking girth rule minm structure possible should have 10 vertices & is connected so its impossible to be disconnected

now using eulu's formula for connected planar graph:-

$$n - m + f = 2$$

$$10 - 15 + f = 2$$

$$\boxed{f = 7}$$

as ginth is 5 so

degree of $\forall$ face ≥ 5

we know that $2m = \sum_{i=1}^{f} d(f_i)$

↳ sum of degree of all face

$$30 = \sum_{i=1}^{f} d(f_i)$$

$$d(f_i) \geq 5$$

doing summation

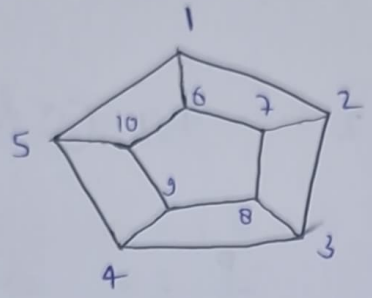
$$\sum_{i=1}^{f} d(f_i) \geq 5f$$

$$30 \geq 5f$$

$$6 \geq f \quad \text{but} \quad f = 7 \text{ contradiction}$$

so no such graph is possible.

Q10 (b):-



$n = 10$, $d(v_i) = 3$ for $i = 1$ to 10

& ~~each cycle~~ is min^m length cycle is

4 (8 - 3 - 2 - 7 - 8)

Q11 :- h = no. of hexagonal faces

p = no. of pentagonal faces

$f = h + p$ —— (1)

↳ no. of faces

n = no. of vertices

m = no. of edges

as it is 3-regular graph so

$$\sum_v \deg(v) = 2m$$

$$\boxed{3 \times n = 2m} \quad \text{——} (2)$$

$$\sum_f \deg(f) = 2m$$

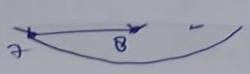
$$\boxed{6h + 5p = 2m} \quad \text{——} (3)$$

using euler's formula

$$\therefore n - m + f = 2$$

$$\frac{2m}{3} - m + f = 2$$

by (2)



$$f - \frac{m}{3} = 2$$

$$(h+p) - \left(\frac{6h+5p}{6}\right) = 2 \qquad \text{by } ① \text{ \& } ③$$

$$h + p - h + \frac{5}{6}p = 2 \qquad \frac{p}{6} = 2 \Rightarrow \boxed{p = 12}$$

by eqn ③

$$2m = 6h + 60$$

by ② $\quad 2m = 3n$

$$3n = 6h + 60$$

$$\boxed{n = 2h + 20}$$ Ans

Soln 12 :- Let G be a simple graph on n vertices, & let G^c be its complement.

As for any simple planar graph on $n \geq 3$ vertices can have at most $3n-6$ edges so

$$m(G) \leq 3n-6 \quad \& \quad m(G^c) \leq 3n-6$$

$$m(G) + m(G^c) = \frac{n \times (n-1)}{2}$$

$$m(G) + m(G^c) \leq 6n-12$$

$$\frac{n \times (n-1)}{2} \leq 6n-12$$

Soln 12:-

$$n^2 - n \leq 12n - 24$$

$$n^2 - 13n + 24 \leq 0$$

for $n^2 - 13n + 24 = 0$ $\quad n = \frac{13 \pm 7.3}{2}$

so $n \approx 2.23$

or $n \approx 10.77$

for $n^2 - 13n + 24 \leq 0$ to be true

$$2.23 \leq n \leq 10.77$$

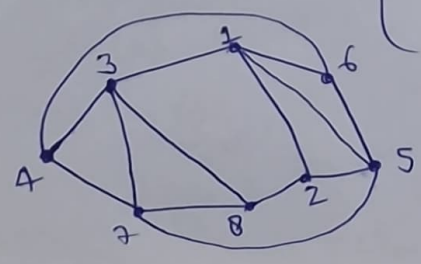
for $n \geq 11$ this inequality fails for K_{11} we need 55 edges but two planar graphs on 11 vertices can give $(3 \times 11 - 6) \times 2 = 54$ edges so for any $n \geq 11$ either G or $\bar{G}$ or both must be non planar.

Part 2:- a graph G on 8 vertices to be self complementary no. of edges $= \frac{1}{2} m(K_8)$

$$m = \frac{1}{2} \times \frac{8 \times 7}{2} = 14$$

so one valid degree seqn :-

$(4, 4, 4, 4, 3, 3, 3, 3)$



it is planar by diagram & ~~th~~ is also self complementary.